1 Heegaard Diagrams of 3-Manifolds

Example 1 ($\mathbf{L(p,q) := S^3/[(w,z) \sim (\xi_p w, \xi_p^q z)]}$)**.** Consider the standard Heegaard decomposition of S^3 given by

$$S^3 = \underbrace{\{|z| \geq |w|\}}_{h^1} \cup \underbrace{|z| = |w|}_{\mathbb{T}^2} \underbrace{\{|w| \geq |z|\}}_{h^2} \tag{1}$$

Then the action of $\mathbb{Z}/(p)$ which I will denote by f preserves the handlebodies in this decomposition alongside the torus $|z| = |w|$. This gives us

$$L(p,q) = h^1/f \cup_{\mathbb{T}^2/f} h^2/f \tag{2}$$

This is useless unless we can describe what these sets look like, notice that if we can describe $\mathbb{T}^2/f$, then we have also described the handlebodies, since these can be expressed as $\mathbb{T}^2/f \times I$. Computing for $\mathbb{T}^2$:

$$\mathbb{T}^2/f = \frac{\mathbb{T}^2}{(x,y) \sim \left(x + \frac{1}{p}, y + \frac{q}{p}\right)} = \frac{\mathbb{R}^2/(\mathbb{Z}^2)}{(\frac{1}{p}, \frac{q}{p})\mathbb{Z}} = \frac{\mathbb{R}^2}{(0,1)\mathbb{Z} \oplus (\frac{1}{p}, \frac{q}{p})\mathbb{Z}} \cong \mathbb{T}^2 \tag{3}$$

Thus equation (2) also specifies a Heegaard Decomposition. Using the basis computed in equation (3) we take the corresponding generators for $H_1(\partial h^j)$. Now $(0,1)\mathbb{Z}$ is the meridian for h^1 and thus also h^1/f, same for $(1,0)$ as the meridian of $h^2, h^2/f$, The meridian of h^2 is thus glued to $p(1/p, q/p) - q(0,1)$. This gives the heegaard decomposition on $\mathbb{T}^2$ as the curves given by $(\alpha, p\beta - q\alpha)$, where $\alpha, \beta \in H^1(\mathbb{T}^2)$ are the homology basis of the quotient computed in equation (3).

Example 2 ($\mathbb{T}^3$ with the flat metric)**.** Model $\mathbb{T}^3$ as $[0, 2\pi)^3$, then consider

$$f(x) = \sum_1^3 \frac{1}{2}\cos(x^j) + \frac{3}{2} \tag{4}$$

I claim that this is a self-indexing morse function, checking this

$$-\nabla f = \frac{1}{2}\sin(x^j)dx_j \implies (-\nabla f)^{-1}(0) = \{(s,t,r) \mid s,t,r \in \{0,\pi\}\} \tag{5}$$

And at such points we have

$$\text{Hess}(f) = \frac{1}{2}\,\text{Diag}(-\cos(x^j)) \tag{6}$$

This verifies f is morse, to see its self indexing note that the index is the number of occurences of 0 in the critical point vector.

Now we solve for a gradient flow of f, from the above computations

$$\dot{x}^j = \frac{1}{2}\sin(x^j) \tag{7}$$

We can apply seperation of variables to $y'/\sin y = \frac{1}{2}$ this gives us $\int \frac{1}{\sin(x^j)} dx^j = \frac{t}{2} + C$ integrating gives us

$$\log(\tan(x^j/2)) = \frac{t}{2} + C \tag{8}$$

and thus

$$x^j = 2\arctan(Ce^{t/2}) \tag{9}$$

Using the initial condition $x^j(0), C = \tan(x^j(0)/2)$ so that

$$x^j = 2\arctan(\tan(x^j(0)/2)e^{t/2}) \tag{10}$$

The Heegaard diagram uses the asymptotic information of gradient flows, luckily this is easy to compute:

$$\lim_{t\to\infty} x^j = \begin{cases} \pi & x^j(0) \neq 0 \\ 0 & x^j(0) = 0 \end{cases} \qquad \lim_{t\to-\infty} x^j = \begin{cases} 0 & x^j(0) \neq \pi \\ \pi & x^j(0) = \pi \end{cases} \tag{11}$$

(We make sense of this by thinking of S^1 as the codomain of tan and domain of arctan) It follows that the stable manifold of the index one critical point $(0, \pi, \pi)$ is given by $\{(0, x^2, x^3) \mid x^2, x^3 \neq 0\}$. The others are described symmetrically. Similarly we can consider the unstable manifold of the critical point $(\pi, 0, 0)$ as $\{(\pi, x^2, x^3) \mid x^2, x^3 \neq \pi\}$, the rest of the cases are symmetric.

Now, to get the Heegaard decomposition, we study the level set $f^{-1}(3/2)$, the α curves are given by

$$\alpha_1 = \{(0, x^2, x^3) \mid \cos(x^2) + \cos(x^3) = -1\} \tag{12}$$
$$\alpha_2 = \{(x^1, 0, x^3) \mid \cos(x^1) + \cos(x^3) = -1\} \tag{13}$$
$$\alpha_3 = \{(x^1, x^2, 0) \mid \cos(x^1) + \cos(x^2) = -1\} \tag{14}$$

And the β curves

$$\beta_1 = \{(\pi, x^2, x^3) \mid \cos(x^2) + \cos(x^3) = 1\} \tag{15}$$
$$\beta_2 = \{(x^1, \pi, x^3) \mid \cos(x^1) + \cos(x^3) = 1\} \tag{16}$$
$$\beta_3 = \{(x^1, x^2, \pi) \mid \cos(x^1) + \cos(x^2) = 1\} \tag{17}$$

2 Symmetric Products

Theorem 1 (Osvath Szabo Intro to Floer 4.1)**.** $\pi_1(\mathrm{Sym}^g(\Sigma)) \cong H_1(\mathrm{Sym}^g(\Sigma)) \cong H_1(\Sigma)$

From Dusa McDuff's survey paper, the isomorphism is induced by inclusion $\Sigma \hookrightarrow \Sigma \times \{z\}^{g-1}$

Exercise 2.0.1. Compute $\pi_2(\mathrm{Sym}^2(\Sigma_2))$

Proof. As written in the notes, $\mathrm{Sym}^2(\Sigma_2) \cong \mathbb{T}^4 \# \mathbb{CP}^2$. Then we know that $\mathbb{CP}^2$ is given by gluing D^4 to $\mathbb{CP}^1 \cong S^2$ along $\partial D^4 = S^3$. We can take the connect sum to be cutting out this D^4 and gluing along S^3. We see that if we first took $(\mathbb{T}^4, D^4)$ and $(\mathbb{CP}^4, D^4)$ as CW pairs and quotiented before gluing then we get $\mathbb{T}^4$ with $\mathbb{CP}^2 \cong S^2$ glued to a point, this is a homotopy equivalence by Hatcher 0.16. Thus we find that

$$\mathrm{Sym}^2(\Sigma_2) \cong \mathbb{T}^4 \vee S^2 \tag{18}$$

Observe that this has universal covering space X given by $\mathbb{R}^4$ with spheres placed at each point in $\mathbb{Z}^4$. The Hurewicz theorem tells us that $\pi_2(X) \cong H_2(X)$, and the fibration

$$\mathbb{Z}^4 \to X \to \mathbb{T}^4 \vee S^2 \tag{19}$$

gives the long exact sequence of homotopy groups (Hatcher 4.2)

$$\cdots \longrightarrow \pi_j(\mathbb{Z}^4) \longrightarrow \pi_j(X) \longrightarrow \pi_j(\mathbb{T}^4 \vee S^2) \longrightarrow \pi_{j-1}(\mathbb{Z}^4) \longrightarrow \cdots \tag{20}$$

Setting $j \geq 2$ the homotopy groups of $\mathbb{Z}^4$ are zero so we get $\pi_2(X) \cong \pi_2(\mathbb{T}^4 \vee S^2)$. Thus it suffices to compute $H_2(X)$, we can see this is $\mathbb{Z}^4$ since $X \simeq \bigvee_{\mathbb{Z}^4} S^2$. We conclude that

$$\pi_2(\mathrm{Sym}^2(\Sigma_2)) \cong \pi_2(\mathbb{T}^4 \vee S^2) \cong \bigoplus_{\mathbb{Z}^4} \mathbb{Z} \tag{21}$$

□

Exercise 2.0.2. Show that $\mathbb{T}_\alpha$ is a totally real submanifold of $\mathrm{Sym}^g(\Sigma)$

Proof. Use abuse of notation to consider $\mathbb{T}_\alpha \subset \Sigma^g$, then since $\mathbb{T}_\alpha$ is a submanifold and so is the diagonal $\Delta = (\Sigma^g)^{S_g}$ and they are disjoint, we can take a tubular neighborhood N of Δ which is disjoint from $\mathbb{T}_\alpha$. Moreover, by possibly shrinking N we may assume that $S_g : N \to N$. It follows that $S_g : \Sigma^g \setminus N \to \Sigma^g \setminus N$ acts properly discontinuously. Consider the following, the horizontal arrows are inclusions and the vertical are quotient by S_g.

$$\begin{array}{ccccc} \mathbb{T}_\alpha & \xrightarrow{\iota} & \Sigma^g \setminus N & \xrightarrow{\iota} & \Sigma^g \\ \downarrow\cong & & \downarrow & & \downarrow \\ \mathbb{T}_\alpha & \xrightarrow{\iota} & \mathrm{Sym}^g(\Sigma) \setminus N & \xrightarrow{\iota} & \mathrm{Sym}^g(\Sigma) \end{array} \tag{22}$$

Apply T_* to the above diagram to get

$$\begin{array}{ccccc} \bigoplus_j T_*\alpha_j & \xrightarrow{\iota} & T_*(\Sigma^g \setminus N) & \xrightarrow{\iota} & \bigoplus_j T_*\Sigma \\ \downarrow\cong & & \downarrow & & \downarrow \\ T_*\mathbb{T}_\alpha & \xrightarrow{\iota} & T_*\,\mathrm{Sym}^g(\Sigma) \setminus N & \xrightarrow{\iota} & T_*\,\mathrm{Sym}^g(\Sigma) \end{array} \tag{23}$$

The complex structure being induced by that of Σ should mean that $J(T_*\mathbb{T}_\alpha) = \bigoplus_\alpha JT$ and that J commutes with the quotient action. This reduces the question of $\mathbb{T}_\alpha$ being totally real to showing that each of α_j being totally real. This is trivial since for $(v, p) \in T_*\alpha_j$ we have Jv is linearly independent from v. □

Exercise 2.0.3. Show that

$$\frac{H_1(\mathrm{Sym}^g(\Sigma))}{\mathbb{T}_\alpha \oplus \mathbb{T}_\beta} \cong \frac{H_1(\Sigma)}{[\alpha_j]_{j=1}^g, [\beta_j]_{j=1}^g} \cong H_1(Y, \mathbb{Z}) \tag{24}$$

Proof. We use that $\iota : \Sigma \hookrightarrow \mathrm{Sym}^g(\Sigma)$ by inclusion onto the first coordinate induces the isomorphism on homology, thus

$$\frac{H_1(\Sigma)}{[\alpha_j]_{j=1}^g, [\beta_j]_{j=1}^g} \cong \frac{H_1(\mathrm{Sym}^g(\Sigma))}{\iota_*[\alpha_j]_{j=1}^g, \iota_*[\beta_j]_{j=1}^g} \tag{25}$$

A generator for $\mathbb{T}_\alpha$ is of the form $\{z\}^{j-1} \times \alpha_j \times \{z\}^{g-j} \simeq \alpha_j \times \{z\}^{g-1} = \iota(\alpha_j)$.

The second isomorphism follows from the Mayer Vietoris sequence

$$H_1(\Sigma) \xrightarrow{\iota} H_1(h^1) \oplus H_1(h^2) \longrightarrow H_1(Y) \longrightarrow H_0(\Sigma) \xrightarrow{\iota} H_0(h^1) \oplus H_0(h^2) \tag{26}$$

Note that the inclusion map on the level of connected components is injective, hence the diagonal map from $H_1(Y)$ is zero, it follows that $H_1(h^1) \oplus H_1(h^2) \twoheadrightarrow H_1(Y)$. Moreover $H_1(Y) \cong H_1(h^1) \oplus H^1(h^2)/\operatorname{Im}\iota$ can be seen geometrically as $H_1(h^1)/[\beta_j]_1^g \oplus H_1(h^2)/[\alpha_j]_1^g \cong \frac{H_1(\Sigma)}{[\alpha_j]_{j=1}^g, [\beta_j]_{j=1}^g}$. □

Exercise 2.0.4. Consider the genus 1 Heegaard diagram $L(p, q)$ with α, β isotoped to have p intersection points. Show that each of the intersection points $\alpha \cap \beta$ have different equivalence classes.

Proof. Work in $\mathbb{R}^2/\mathbb{Z}^2$ with b the curve having slope $-p/q$. The cosets for intersection points are thus $(k(q/p), 0)$ for $k = 0, \ldots, p-1$. Then taking two intersection points, $(k(q/p), 0)$ and $(r(q/p), 0)$ where we may assume $k > r$ We find that the curve a connecting them lies in $\mathbb{R} \times 0$, and we may choose the curve b to be $b(t) = t\frac{(k-r)}{p}(q, p)$ It follows that $[b - a]$ corresponds to $(0, k - r) \in \mathbb{R}^2$ But $(0, k - r) \neq 0 \in \frac{\mathbb{R}^2}{\mathbb{R}\times\{0\}\oplus\mathbb{R}(q,p)} \cong H_1(\Sigma)/[\alpha] + [\beta]$ □